Full Roadmap Tracker — Java + Spring Boot + HLD/System Design

Off: Sat & Sun

Har phase ke start pe resource link diya hai, aur har row mein completion ke waqt jo git commit karna hai wo already likha hua hai — bas us din ka kaam khatam karke wahi commit message use kar dena.

PHASE 0 — Java Full Course (CoderArmy, 57 videos)

Pace: 2 videos/day | 29 working days

Resource: https://www.youtube.com/playlist?list=PLQEaRBV9gAFsR15tNo2QLF9d2qc-c018p

(Note: source doc mein sirf HLD phase ke commits explicitly diye the. Java/Spring Boot ke liye same convention pe commit names bana diye hain — agar tum koi aur naming chahte ho to bata dena.)

Day	Videos	Commit (on completion)
1	1–2	`java-day-01-videos-1-2`
2	3–4	`java-day-02-videos-3-4`
3	5–6	`java-day-03-videos-5-6`
4	7–8	`java-day-04-videos-7-8`
5	9–10	`java-day-05-videos-9-10`
6	11–12	`java-day-06-videos-11-12`
7	13–14	`java-day-07-videos-13-14`
8	15–16	`java-day-08-videos-15-16`
9	17–18	`java-day-09-videos-17-18`
10	19–20	`java-day-10-videos-19-20`
11	21–22	`java-day-11-videos-21-22`
12	23–24	`java-day-12-videos-23-24`
13	25–26	`java-day-13-videos-25-26`
14	27–28	`java-day-14-videos-27-28`
15	29–30	`java-day-15-videos-29-30`
16	31–32	`java-day-16-videos-31-32`

Day	Videos	Commit (on completion)
17	33–34	`java-day-17-videos-33-34`
18	35–36	`java-day-18-videos-35-36`
19	37–38	`java-day-19-videos-37-38`
20	39–40	`java-day-20-videos-39-40`
21	41–42	`java-day-21-videos-41-42`
22	43–44	`java-day-22-videos-43-44`
23	45–46	`java-day-23-videos-45-46`
24	47–48	`java-day-24-videos-47-48`
25	49–50	`java-day-25-videos-49-50`
26	51–52	`java-day-26-videos-51-52`
27	53–54	`java-day-27-videos-53-54`
28	55–56	`java-day-28-videos-55-56`
29	57 (last video)	`java-day-29-video-57-final`

PHASE 0.5 — Spring Boot Full Course (CoderArmy, 41 videos)

Pace: 2 videos/day | 21 working days

Resource: https://www.youtube.com/playlist?list=PLEYgx5hMdopw *(source doc mein ye playlist ID incomplete lag rahi thi — link confirm/verify kar lena before starting.)*

Day	Videos	Commit (on completion)
1	1–2	`spring-day-01-videos-1-2`
2	3–4	`spring-day-02-videos-3-4`
3	5–6	`spring-day-03-videos-5-6`
4	7–8	`spring-day-04-videos-7-8`
5	9–10	`spring-day-05-videos-9-10`
6	11–12	`spring-day-06-videos-11-12`
7	13–14	`spring-day-07-videos-13-14`
8	15–16	`spring-day-08-videos-15-16`
9	17–18	`spring-day-09-videos-17-18`
10	19–20	`spring-day-10-videos-19-20`
11	21–22	`spring-day-11-videos-21-22`
12	23–24	`spring-day-12-videos-23-24`
13	25–26	`spring-day-13-videos-25-26`
14	27–28	`spring-day-14-videos-27-28`
15	29–30	`spring-day-15-videos-29-30`
16	31–32	`spring-day-16-videos-31-32`

Day	Videos	Commit (on completion)
17	33–34	`spring-day-17-videos-33-34`
18	35–36	`spring-day-18-videos-35-36`
19	37–38	`spring-day-19-videos-37-38`
20	39–40	`spring-day-20-videos-39-40`
21	41 (last video)	`spring-day-21-video-41-final`

(Spring Boot khatam hote hi agle working day se seedha HLD start.)

PHASE 1 — High Level Design (HLD) — 2 Hours/Day

48 working days | One system per day rule

Primary resources (all weeks):

- donnemartin/system-design-primer — https://github.com/donnemartin/system-design-primer
- ashishps1/awesome-system-design-resources — https://github.com/ashishps1/awesome-system-design-resources

Full 15-repo list is in the Reference Section at the end of this file.

Week 1 — Foundations + Core Concepts (Target: Distributed Session Store)

Resources: system-design-primer, alex/what-happens-when — https://github.com/alex/what-happens-when

Day	Topic	Commit (on completion)
1	Networking Foundations	`hld-day-01-networking-foundations`
2	Scalability/Availability/Reliability/SPOF	`hld-day-02-scalability-availability`
3	CAP Theorem + PACELC	`hld-day-03-cap-pacelc`
4	Session Store: Consistency + Availability	`hld-day-04-session-store-consistency-availability`
5	Session Store: Hashing + Partition Challenge	`hld-day-05-session-store-hashing-partition`
6	Session Store: Fault Tolerance + Deliverable	`hld-day-06-session-store-deliverable`

Week 2 — Networking & API Layer (Target: Ride-Hailing App API)

Resources: system-design-primer, ashishps1, Chalarangelo/30-seconds-of-code — https://github.com/Chalarangelo/30-seconds-of-code, CodingChallengesFYI/SharedSolutions — https://github.com/CodingChallengesFYI/SharedSolutions

Day	Topic	Commit (on completion)
7	DNS + Load Balancing	`hld-day-07-dns-loadbalancing`
8	Reverse Proxy + API Gateway	`hld-day-08-reverse-proxy-gateway`
9	CDN	`hld-day-09-cdn`
10	Ride API: Contract + Style Decision	`hld-day-10-ride-api-contract`
11	Ride API: Real-Time Tracking	`hld-day-11-ride-api-realtime`
12	Ride API: Idempotency + Rate Limiter + Deliverable	`hld-day-12-ride-api-deliverable`

Week 3 — Database Layer (Target: Ride-Tracking App DB)

Resources: system-design-primer, ashishps1, awesome-scalability — https://github.com/binhnguyennus/awesome-scalability

Day	Topic	Commit (on completion)
13	ACID + SQL vs NoSQL	`hld-day-13-acid-sql-nosql`
14	Database Indexes + Bloom Filters	`hld-day-14-indexes-bloom-filters`
15	Replication	`hld-day-15-replication`
16	Ride DB: Sharding + Federation	`hld-day-16-ride-db-sharding-federation`
17	Ride DB: Denormalization + Tuning Challenge	`hld-day-17-ride-db-denormalization-tuning`
18	Ride DB: Architecture + Scaling Plan + Deliverable	`hld-day-18-ride-db-deliverable`

Week 4 — Caching Layer (Target: Product Catalog Caching Layer)

Resources: ashishps1, InterviewReady/system-design-resources — https://github.com/InterviewReady/system-design-resources, awesome-scalability, CodingChallengesFYI/SharedSolutions

Day	Topic	Commit (on completion)
19	Caching 101 + Strategies	`hld-day-19-caching-strategies`
20	Cache Eviction Policies	`hld-day-20-cache-eviction`
21	Distributed Caching	`hld-day-21-distributed-caching`
22	Catalog: Cache Strategy + CDN Split	`hld-day-22-catalog-cache-strategy`
23	Catalog: Case Study + Build Challenge	`hld-day-23-catalog-case-study-build`
24	Catalog: Invalidation + Stampede + Deliverable	`hld-day-24-catalog-caching-deliverable`

Week 5 — Async, Messaging & Distributed Systems (Target: Order-Processing Async Pipeline)

Resources: ashishps1, InterviewReady/system-design-resources, arpit20adlakha/Computer-Science-Papers-For-System-Design — https://github.com/arpit20adlakha/Computer-Science-Papers-For-System-Design

Day	Topic	Commit (on completion)
25	Message Queues + Pub/Sub	`hld-day-25-mq-pubsub`
26	Change Data Capture + Event-Driven Basics	`hld-day-26-cdc-event-driven`
27	Heartbeats + Service Discovery	`hld-day-27-heartbeats-service-discovery`
28	Order Pipeline: Queue Structure + Leader Election	`hld-day-28-order-pipeline-queue-consensus`
29	Order Pipeline: Locking + Consensus Challenge	`hld-day-29-order-pipeline-locking-consensus`
30	Order Pipeline: Gossip + Circuit Breaker + Deliverable	`hld-day-30-order-pipeline-deliverable`

Week 6 — Architecture Patterns + Tradeoffs (Target: any 1 sample system)

Resources: ashishps1, addyosmani/essential-js-design-patterns — https://github.com/addyosmani/essential-js-design-patterns

Day	Topic	Commit (on completion)
31	Client-Server + Microservices Architecture	`hld-day-31-client-server-microservices`
32	Serverless + Event-Driven Architecture	`hld-day-32-serverless-event-driven`
33	P2P Architecture + Pattern Selection	`hld-day-33-p2p-pattern-selection`
34	Tradeoffs Part 1	`hld-day-34-tradeoffs-1`
35	Tradeoffs Part 2	`hld-day-35-tradeoffs-2`
36	Tradeoffs Part 3 + Deliverable	`hld-day-36-tradeoffs-deliverable`

Week 7 — Easy Interview Problems, Hands-On (full interview format daily)

Resources: system-design-primer, ashishps1, 30-seconds-of-code, InterviewReady/system-design-resources

Day	Topic	Commit (on completion)
37	Design URL Shortener (TinyURL)	`hld-day-37-url-shortener`
38	Design Rate Limiter (full interview format)	`hld-day-38-rate-limiter-full-design`
39	Design Parking Garage	`hld-day-39-parking-garage-scale`
40	Design Distributed Key-Value Store / Cache	`hld-day-40-distributed-kv-store`
41	Design Authentication System	`hld-day-41-authentication-system`
42	Design Web Crawler (Review Day)	`hld-day-42-web-crawler-review`

Week 8 — Medium + Hard, Capstone

Resources: ashishps1 Medium/Hard list, system-design-primer, design-guru/grokking-system-design (verify exact repo URL yourself, source doc naming was slightly inconsistent), yangshun/tech-interview-handbook — https://github.com/yangshun/tech-interview-handbook

Day	Topic	Commit (on completion)
43	Design WhatsApp / Chat App	`hld-day-43-whatsapp`
44	Design Instagram/Twitter Feed	`hld-day-44-social-feed`
45	Design Netflix / YouTube	`hld-day-45-video-streaming`
46	Design Notification & Scheduling System	`hld-day-46-notification-job-scheduler`
47	Design Uber — Part 1 (Capstone)	`hld-day-47-uber-capstone-draft`
48	Design Uber — Part 2, Capstone Finalize	`hld-day-48-capstone-final`

Reference Section — Full Repo List (all 15, for HLD phase)

1. design-guru/grokking-system-design
2. yangshun/tech-interview-handbook
3. kdn251/interviews
4. alex/what-happens-when
5. Chalarangelo/30-seconds-of-code
6. codersguild/System-Design
7. arpit20adlakha/Computer-Science-Papers-For-System-Design
8. InterviewReady/system-design-resources
9. arpit20adlakha/Data-Structure-Algorithms-LLD-HLD
10. kilimchoi/engineering-blogs
11. donnemartin/system-design-primer

12. addyosmani/essential-js-design-patterns
13. binhnguyennus/awesome-scalability
14. ashishps1/awesome-system-design-resources
15. CodingChallengesFYI/SharedSolutions

Tip: is file ko apne roadmap-tracker Streamlit app ke sath ya manually use kar sakte ho — jis din ka kaam ho jaye, us row ka commit message copy karke wahi commit kar dena.